

Number Conversion & Binary Arithmetic — Solutions with Step-by-step Workings

1. Convert binary number 110110101 to decimal.

Method: Sum $\text{bit} \times 2^{\text{position}}$. Terms:

$$1 \times 2^8 = 1 \times 256 = 256$$

$$1 \times 2^7 = 1 \times 128 = 128$$

$$0 \times 2^6 = 0 \times 64 = 0$$

$$1 \times 2^5 = 1 \times 32 = 32$$

$$1 \times 2^4 = 1 \times 16 = 16$$

$$0 \times 2^3 = 0 \times 8 = 0$$

$$1 \times 2^2 = 1 \times 4 = 4$$

$$0 \times 2^1 = 0 \times 2 = 0$$

$$1 \times 2^0 = 1 \times 1 = 1$$

Answer: 437 (decimal)

2. Convert decimal number 987 to binary.

Method: Repeated division by 2. Steps:

$$493 \div 2 = 493 \text{ remainder } 1$$

$$246 \div 2 = 246 \text{ remainder } 1$$

$$123 \div 2 = 123 \text{ remainder } 0$$

$$61 \div 2 = 61 \text{ remainder } 1$$

$$30 \div 2 = 30 \text{ remainder } 1$$

$$15 \div 2 = 15 \text{ remainder } 0$$

$$7 \div 2 = 7 \text{ remainder } 1$$

$$3 \div 2 = 3 \text{ remainder } 1$$

$$1 \div 2 = 1 \text{ remainder } 1$$

$$0 \div 2 = 0 \text{ remainder } 1$$

Answer: 1111011011 (binary)

3. Convert hexadecimal number 3A7 to decimal.

Method: Hex digits $\times 16^{\text{position}}$. Terms:

$$3 \times 16^2 = 3 \times 256 = 768$$

$$10 \times 16^1 = 10 \times 16 = 160$$

$$7 \times 16^0 = 7 \times 1 = 7$$

Answer: 935 (decimal)

4. Convert binary number 10011011101 to hexadecimal.

Method: Group binary in 4s from right:

$$0100 \rightarrow 4$$

$$1101 \rightarrow 13$$

$$1101 \rightarrow 13$$

Answer: 4DD (hexadecimal)

5. Convert octal number 765 to binary.

Method: Convert each octal digit to 3-bit binary:

$$7 \rightarrow 111$$

$$6 \rightarrow 110$$

$$5 \rightarrow 101$$

Answer: 111110101 (binary)

6. Convert binary number 111011101101 to octal.

Method: Group binary in 3s from right:

$$111 \rightarrow 7$$

$$011 \rightarrow 3$$

$$101 \rightarrow 5$$

$$101 \rightarrow 5$$

Answer: 7355 (octal)

7. Convert hexadecimal number 1F4 to binary.

Method: Convert each hex digit to 4-bit binary:

$$1 \rightarrow 0001$$

$$F \rightarrow 1111$$

$$4 \rightarrow 0100$$

Answer: 000111110100 (binary)

8. Convert binary number 101110010 to hexadecimal.

Method: Group binary in 4s from right:

0001 -> 1

0111 -> 7

0010 -> 2

Answer: 172 (hexadecimal)

9. Convert decimal number 2047 to binary.

Method: Repeated division by 2. Steps:

$1023 \div 2 = 1023$ remainder 1

$511 \div 2 = 511$ remainder 1

$255 \div 2 = 255$ remainder 1

$127 \div 2 = 127$ remainder 1

$63 \div 2 = 63$ remainder 1

$31 \div 2 = 31$ remainder 1

$15 \div 2 = 15$ remainder 1

$7 \div 2 = 7$ remainder 1

$3 \div 2 = 3$ remainder 1

$1 \div 2 = 1$ remainder 1

$0 \div 2 = 0$ remainder 1

Answer: 1111111111 (binary)

10. Convert binary number 11111010011 to decimal.

Method: Sum $\text{bit} \times 2^{\text{position}}$. Terms:

$1 \times 2^{10} = 1 \times 1024 = 1024$

$1 \times 2^9 = 1 \times 512 = 512$

$1 \times 2^8 = 1 \times 256 = 256$

$1 \times 2^7 = 1 \times 128 = 128$

$1 \times 2^6 = 1 \times 64 = 64$

$0 \times 2^5 = 0 \times 32 = 0$

$1 \times 2^4 = 1 \times 16 = 16$

$0 \times 2^3 = 0 \times 8 = 0$

$0 \times 2^2 = 0 \times 4 = 0$

$1 \times 2^1 = 1 \times 2 = 2$

$1 \times 2^0 = 1 \times 1 = 1$

Answer: 2003 (decimal)

11. Convert hexadecimal number 4B2 to octal.

Method: Hex -> Decimal -> Octal. Hex to decimal terms:

$$4 \times 16^2 = 4 \times 256 = 1024$$

$$11 \times 16^1 = 11 \times 16 = 176$$

$$2 \times 16^0 = 2 \times 1 = 2$$

Then decimal 1202 to octal by repeated division. Answer: 2262 (octal)

12. Convert octal number 777 to hexadecimal.

Method: Octal -> Decimal -> Hex. Octal to decimal terms:

$$7 \times 8^2 = 7 \times 64 = 448$$

$$7 \times 8^1 = 7 \times 8 = 56$$

$$7 \times 8^0 = 7 \times 1 = 7$$

Then decimal 511 to hex by repeated division. Answer: 1FF (hexadecimal)

13. Convert binary number 111000111011 to decimal.

Method: Sum $\text{bit} \times 2^{\text{position}}$. Terms:

$$1 \times 2^{11} = 1 \times 2048 = 2048$$

$$1 \times 2^{10} = 1 \times 1024 = 1024$$

$$1 \times 2^9 = 1 \times 512 = 512$$

$$0 \times 2^8 = 0 \times 256 = 0$$

$$0 \times 2^7 = 0 \times 128 = 0$$

$$0 \times 2^6 = 0 \times 64 = 0$$

$$1 \times 2^5 = 1 \times 32 = 32$$

$$1 \times 2^4 = 1 \times 16 = 16$$

$$1 \times 2^3 = 1 \times 8 = 8$$

$$0 \times 2^2 = 0 \times 4 = 0$$

$$1 \times 2^1 = 1 \times 2 = 2$$

$$1 \times 2^0 = 1 \times 1 = 1$$

Answer: 3643 (decimal)

14. Convert octal number 320 to binary.

Method: Convert each octal digit to 3-bit binary:

$$3 \rightarrow 011$$

$$2 \rightarrow 010$$

$$0 \rightarrow 000$$

Answer: 011010000 (binary)

15. Convert decimal number 4095 to hexadecimal.

Method: Repeated division by 16. Steps:

$$255 \div 16 = 15 \text{ remainder } (F)$$

$$15 \div 16 = 15 \text{ remainder } (F)$$

$$0 \div 16 = 0 \text{ remainder } (F)$$

Answer: FFF (hexadecimal)

16. Convert hexadecimal number 2A3F to binary.

Method: Convert each hex digit to 4-bit binary:

$$2 \rightarrow 0010$$

$$A \rightarrow 1010$$

$$3 \rightarrow 0011$$

$$F \rightarrow 1111$$

Answer: 0010101000111111 (binary)

17. Convert binary number 10000000 to octal.

Method: Group binary in 3s from right:

$$100 \rightarrow 4$$

$$000 \rightarrow 0$$

$$000 \rightarrow 0$$

Answer: 400 (octal)

18. Convert octal number 377 to binary.

Method: Convert each octal digit to 3-bit binary:

$$3 \rightarrow 011$$

$$7 \rightarrow 111$$

$$7 \rightarrow 111$$

Answer: 011111111 (binary)

19. Convert hexadecimal number 7D to decimal.

Method: Hex digits $\times 16^{\text{position}}$. Terms:

$$7 \times 16^1 = 7 \times 16 = 112$$

$$13 \times 16^0 = 13 \times 1 = 13$$

Answer: 125 (decimal)

20. Convert binary number 100110101110 to octal.

Method: Group binary in 3s from right:

100 -> 4

110 -> 6

101 -> 5

110 -> 6

Answer: 4656 (octal)

21. Convert decimal number 555 to hexadecimal.

Method: Repeated division by 16. Steps:

$$34 \div 16 = 2 \text{ remainder } 2$$

$$2 \div 16 = 0 \text{ remainder } 2$$

$$0 \div 16 = 0 \text{ remainder } 0$$

Answer: 22B (hexadecimal)

22. Convert hexadecimal number 1E240 to decimal.

Method: Hex digits $\times 16^{\text{position}}$. Terms:

$$1 \times 16^4 = 1 \times 65536 = 65536$$

$$14 \times 16^3 = 14 \times 4096 = 57344$$

$$2 \times 16^2 = 2 \times 256 = 512$$

$$4 \times 16^1 = 4 \times 16 = 64$$

$$0 \times 16^0 = 0 \times 1 = 0$$

Answer: 123456 (decimal)

23. Convert binary number 1111111111 to hexadecimal.

Method: Group binary in 4s from right:

$$1111 \rightarrow 15$$

$$1111 \rightarrow 15$$

$$1111 \rightarrow 15$$

Answer: FFF (hexadecimal)

24. Convert decimal number 1023 to octal.

Method: Repeated division by 8. Steps:

$$127 \div 8 = 15 \text{ remainder } 7$$

$$15 \div 8 = 1 \text{ remainder } 7$$

$$1 \div 8 = 0 \text{ remainder } 1$$

$$0 \div 8 = 0 \text{ remainder } 0$$

Answer: 1777 (octal)

25. Convert binary number 101010101 to decimal.

Method: Sum $\text{bit} \times 2^{\text{position}}$. Terms:

$$1 \times 2^8 = 1 \times 256 = 256$$

$$0 \times 2^7 = 0 \times 128 = 0$$

$$1 \times 2^6 = 1 \times 64 = 64$$

$$0 \times 2^5 = 0 \times 32 = 0$$

$$1 \times 2^4 = 1 \times 16 = 16$$

$$0 \times 2^3 = 0 \times 8 = 0$$

$$1 \times 2^2 = 1 \times 4 = 4$$

$$0 \times 2^1 = 0 \times 2 = 0$$

$$1 \times 2^0 = 1 \times 1 = 1$$

Answer: 341 (decimal)

26. Convert hexadecimal number 3F2 to binary.

Method: Convert each hex digit to 4-bit binary:

$$3 \rightarrow 0011$$

$$F \rightarrow 1111$$

$$2 \rightarrow 0010$$

Answer: 001111110010 (binary)

27. Convert octal number 12345 to decimal.

Method: Octal digits $\times 8^{\text{position}}$. Terms:

$$1 \times 8^4 = 1 \times 4096 = 4096$$

$$2 \times 8^3 = 2 \times 512 = 1024$$

$$3 \times 8^2 = 3 \times 64 = 192$$

$$4 \times 8^1 = 4 \times 8 = 32$$

$$5 \times 8^0 = 5 \times 1 = 5$$

Answer: 5349 (decimal)

28. Convert binary number 100110010011 to hexadecimal.

Method: Group binary in 4s from right:

$$1001 \rightarrow 9$$

$$1001 \rightarrow 9$$

$$0011 \rightarrow 3$$

Answer: 993 (hexadecimal)

29. Convert hexadecimal number ABCD to binary.

Method: Convert each hex digit to 4-bit binary:

A -> 1010

B -> 1011

C -> 1100

D -> 1101

Answer: 1010101111001101 (binary)

30. Convert decimal number 999 to hexadecimal.

Method: Repeated division by 16. Steps:

$62 \div 16 = 3$ remainder 7

$3 \div 16 = 0$ remainder (E)

$0 \div 16 = 0$ remainder 3

Answer: 3E7 (hexadecimal)

Binary Addition — Step-by-step

31. Add binary numbers 1011 and 1101.

Binary to decimal: 1011 -> 11, 1101 -> 13

Step-by-step column addition (right to left):

Adding bits $1 + 1 + \text{carry } 0 = 2 \rightarrow$ result bit 0, new carry 1

Adding bits $1 + 0 + \text{carry } 1 = 2 \rightarrow$ result bit 0, new carry 1

Adding bits $0 + 1 + \text{carry } 1 = 2 \rightarrow$ result bit 0, new carry 1

Adding bits $1 + 1 + \text{carry } 1 = 3 \rightarrow$ result bit 1, new carry 1

Final carry 1 appended.

Result: 11000 (binary) = 24 (decimal)

32. Add binary numbers 11101 and 10110.

Binary to decimal: 11101 -> 29, 10110 -> 22

Step-by-step column addition (right to left):

Adding bits $1 + 0 + \text{carry } 0 = 1 \rightarrow$ result bit 1, new carry 0

Adding bits $0 + 1 + \text{carry } 0 = 1 \rightarrow$ result bit 1, new carry 0

Adding bits $1 + 1 + \text{carry } 0 = 2 \rightarrow$ result bit 0, new carry 1

Adding bits $1 + 0 + \text{carry } 1 = 2 \rightarrow$ result bit 0, new carry 1

Adding bits $1 + 1 + \text{carry } 1 = 3 \rightarrow$ result bit 1, new carry 1

Final carry 1 appended.

Result: 110011 (binary) = 51 (decimal)

33. Add binary numbers 100111 and 111001.

Binary to decimal: 100111 -> 39, 111001 -> 57

Step-by-step column addition (right to left):

Adding bits $1 + 1 + \text{carry } 0 = 2 \rightarrow$ result bit 0, new carry 1

Adding bits $1 + 0 + \text{carry } 1 = 2 \rightarrow$ result bit 0, new carry 1

Adding bits $1 + 0 + \text{carry } 1 = 2 \rightarrow$ result bit 0, new carry 1

Adding bits $0 + 1 + \text{carry } 1 = 2 \rightarrow$ result bit 0, new carry 1

Adding bits $0 + 1 + \text{carry } 1 = 2 \rightarrow$ result bit 0, new carry 1

Adding bits $1 + 1 + \text{carry } 1 = 3 \rightarrow$ result bit 1, new carry 1

Final carry 1 appended.

Result: 1100000 (binary) = 96 (decimal)

34. Add binary numbers 110011 and 1001.

Binary to decimal: 110011 -> 51, 1001 -> 9

Step-by-step column addition (right to left):

Adding bits $1 + 1 + \text{carry } 0 = 2 \rightarrow$ result bit 0, new carry 1

Adding bits $1 + 0 + \text{carry } 1 = 2 \rightarrow$ result bit 0, new carry 1

Adding bits $0 + 0 + \text{carry } 1 = 1 \rightarrow$ result bit 1, new carry 0

Adding bits $0 + 1 + \text{carry } 0 = 1 \rightarrow$ result bit 1, new carry 0

Adding bits $1 + 0 + \text{carry } 0 = 1 \rightarrow$ result bit 1, new carry 0

Adding bits $1 + 0 + \text{carry } 0 = 1 \rightarrow$ result bit 1, new carry 0

Result: 111100 (binary) = 60 (decimal)

35. Add binary numbers 1010101 and 110110.

Binary to decimal: 1010101 -> 85, 110110 -> 54

Step-by-step column addition (right to left):

Adding bits $1 + 0 + \text{carry } 0 = 1 \rightarrow$ result bit 1, new carry 0

Adding bits $0 + 1 + \text{carry } 0 = 1 \rightarrow$ result bit 1, new carry 0

Adding bits $1 + 1 + \text{carry } 0 = 2 \rightarrow$ result bit 0, new carry 1

Adding bits $0 + 0 + \text{carry } 1 = 1 \rightarrow$ result bit 1, new carry 0

Adding bits $1 + 1 + \text{carry } 0 = 2 \rightarrow$ result bit 0, new carry 1

Adding bits $0 + 1 + \text{carry } 1 = 2 \rightarrow$ result bit 0, new carry 1

Adding bits $1 + 0 + \text{carry } 1 = 2 \rightarrow$ result bit 0, new carry 1

Final carry 1 appended.

Result: 10001011 (binary) = 139 (decimal)

Binary Subtraction — Step-by-step

36. Subtract binary number 1011 from 11010.

Binary to decimal: 11010 -> 26, 1011 -> 11

Step-by-step (right to left):

Position 4: need borrow. (after borrow da=2, db=1) -> result bit 1, new borrow 1

Position 3: need borrow. (after borrow da=2, db=1) -> result bit 1, new borrow 1

Position 2: need borrow. (after borrow da=1, db=0) -> result bit 1, new borrow 1

Position 1: need borrow. (after borrow da=2, db=1) -> result bit 1, new borrow 1

Position 0: no borrow. da=0, db=0 -> result bit 0, new borrow 0

Result: 1111 (binary) = 15 (decimal)

37. Subtract binary number 1101 from 101101.

Binary to decimal: 101101 -> 45, 1101 -> 13

Step-by-step (right to left):

Position 5: no borrow. da=1, db=1 -> result bit 0, new borrow 0

Position 4: no borrow. da=0, db=0 -> result bit 0, new borrow 0

Position 3: no borrow. da=1, db=1 -> result bit 0, new borrow 0

Position 2: no borrow. da=1, db=1 -> result bit 0, new borrow 0

Position 1: no borrow. da=0, db=0 -> result bit 0, new borrow 0

Position 0: no borrow. da=1, db=0 -> result bit 1, new borrow 0

Result: 100000 (binary) = 32 (decimal)

38. Subtract binary number 101011 from 111000.

Binary to decimal: 111000 -> 56, 101011 -> 43

Step-by-step (right to left):

Position 5: need borrow. (after borrow da=2, db=1) -> result bit 1, new borrow 1

Position 4: need borrow. (after borrow da=1, db=1) -> result bit 0, new borrow 1

Position 3: need borrow. (after borrow da=1, db=0) -> result bit 1, new borrow 1

Position 2: need borrow. (after borrow da=2, db=1) -> result bit 1, new borrow 1

Position 1: no borrow. da=0, db=0 -> result bit 0, new borrow 0

Position 0: no borrow. da=1, db=1 -> result bit 0, new borrow 0

Result: 1101 (binary) = 13 (decimal)

39. Subtract binary number 1111 from 100000.

Binary to decimal: 100000 -> 32, 1111 -> 15

Step-by-step (right to left):

Position 5: need borrow. (after borrow da=2, db=1) -> result bit 1, new borrow 1

Position 4: need borrow. (after borrow da=1, db=1) -> result bit 0, new borrow 1

Position 3: need borrow. (after borrow da=1, db=1) -> result bit 0, new borrow 1

Position 2: need borrow. (after borrow da=1, db=1) -> result bit 0, new borrow 1

Position 1: need borrow. (after borrow da=1, db=0) -> result bit 1, new borrow 1

Position 0: no borrow. da=0, db=0 -> result bit 0, new borrow 0

Result: 10001 (binary) = 17 (decimal)

40. Subtract binary number 101010 from 1100111.

Binary to decimal: 1100111 -> 103, 101010 -> 42

Step-by-step (right to left):

Position 6: no borrow. da=1, db=0 -> result bit 1, new borrow 0

Position 5: no borrow. da=1, db=1 -> result bit 0, new borrow 0

Position 4: no borrow. da=1, db=0 -> result bit 1, new borrow 0

Position 3: need borrow. (after borrow da=2, db=1) -> result bit 1, new borrow 1

Position 2: need borrow. (after borrow da=1, db=0) -> result bit 1, new borrow 1

Position 1: need borrow. (after borrow da=2, db=1) -> result bit 1, new borrow 1

Position 0: no borrow. da=0, db=0 -> result bit 0, new borrow 0

Result: 111101 (binary) = 61 (decimal)